

FAB Group Project

Weekly Module 1: Search

Objective | Define and implement: (1) a **goal state**; (2) a **set of operators** (actions), and (3) a set of **heuristics** (if-then rules regarding when to use actions). Note that your definition of your goal state will also serve as your basic **evaluation criteria** for assigning credit.

Resources | For now, your “initial state” should be only PDFs of Course Notes covered so far (Credit Assignment, Info. Proc., Search). Later on, you will add many other sources.

Current gap | Current AI tools can support studying by easily generating summaries of any texts of books or articles. Yet there are at least two challenges:

- The space of *plausible* summaries is vast. Current tools do not always know how to evaluate which summary would be most relevant to you at a particular moment.
- Users want summaries of the *same length*, regardless of the length of the source text. Hence, there are limits to *compressing* information.

What is needed is support for the *process* of using AI-generated summaries to “search” for an understanding of the text. This is the idea of **means-ends analysis** from our course: to iteratively reduce the difference between an initial state of confusion, and a goal state of understanding.

Recommended Steps | We recommend proceeding in the following order:

0. Make sure you understand the concepts of states, operators, and heuristics from the course notes on search.

1. Discuss the actions that you as students usually take when studying the course notes, such as “Skim the text”, or “Ask GPT for a summary”. Try to be as specific as possible and write down as many as possible. You can revise and refine these later. Ultimately, you should define your actions precisely as “**operators**” that are part of your search program.

2. Discuss the strategies that you as students use to make your studying efficient. These will be your “**heuristics**”, and should correspond to *when* you take certain actions. They should ultimately be made precise by defining them as “if-then” rules, meaning “if” you are in X situation, then you take Y action. For example, “If I’m just starting to study the course notes, the first thing I do is ask for a general summary”.

3. Define your **goal state**: what does the “output” look like from using your operators and heuristics successfully? Is it a set of summaries? A diagram? Something else? Be as specific as possible.

4. Implement your operators, heuristics, and goal state as a **program** (plan), and test it out on one of the Course Notes PDFs until you get an output that you are satisfied with.

5. Take a **screenshot** of a goal state, write a **short paragraph** describing what you have done, and **upload on KLMS**.